Chih-Wei Joshua Liu

+1 (617) 580-7953 · cjoshliu@stanford.edu · GitHub: cjoshliu

EDUCATION

Stanford University — Stanford, CA, USA
Biophysics PhD Program · **Thesis advisor:** Grant M. Rotskoff · **GPA:** 4.0/4.0 — 09/2024 – present
Courses: CS229 Machine Learning, STATS217 Stochastic Processes, MATH220A Partial Differential Equations

Massachusetts Institute of Technology (MIT) — Cambridge, MA, USA
Bachelor of Science · **Major:** Physics · **GPA:** 4.9/5.0 · **MCAT:** 528/528 — 05/2022
Courses: 18.650 Statistics, 18.354 Nonlinear Dynamics · **Honors:** Joel Matthew Orloff Award, MIT News profile

National Experimental High School — Hsinchu, Taiwan, ROC
SAT: 1600/1600 · **ACT:** 36/36 · **USABO:** High Honorist (2017); Semifinalist (2015 – 2018) — 05/2018

EXPERIENCE

MIT Physics of Living and Nonequilibrium Systems Group — 06/2022 – 08/2024
Research Assistant — Cambridge, MA, USA
Designed polyspectral, ensemble-learning, and neural-network analyses of biophysical time series.

MIT Department of Physics — 02/2024 – 05/2024
Teaching Assistant, 8.241 Biological Physics — Cambridge, MA, USA
Led recitations on fluctuation-dissipation theorems, diffusions, and random walks.

Massachusetts General Hospital — 06/2021 – 08/2021
Data Science Intern — Boston, MA, USA
Implemented GLS models of patient data and authored two clinical publications using `statsmodels`, `R`, and `SPSS`

SELECTED PROJECTS

C.-W.J. Liu, J. Klinger, G.M. Rotskoff, **Optimal parameterization of nonequilibrium generalized master equations from discrete-time experimental data**, *Phys. Rev. X*, provisionally accepted (2026). [code]

- Derived the first maximum-likelihood parameterization of non-Markovian master equations satisfying all physical constraints in general stochastic processes. Validated on three experimental datasets.
- Set new state-of-the-art in benchmark extraction of HP35-Lys24Nle/Asn27His/Lys29Nle protein-folding kinetics from coarse-grained models trained on atomistic molecular-dynamics simulation.
- **Skills:** convex optimization, vector autoregression, model selection, `uv`, `ruff`, `ty`, `lightning`

J. Li*, **C.-W.J. Liu***, M. Szurek, and N. Fakhri, **Measuring irreversibility from learned representations of biological patterns**, *PRX Life* **2**, 033013 (2024), *equal contribution. [code]

- Extended thermodynamic-irreversibility estimation to entire simulated and experimental videos of high-dimensional biological processes using variational-autoencoder representations. **Skills:** numerical integration, `torch`

H. Lee*, E. Koskelo*, S. Gokhale*, J. Li*, C. Fei, **C.-W.J. Liu**, L. Lin, J. Dunkel, D.J. Skinner, and N. Fakhri, **Topological flowscape reveals state transitions in nonreciprocal living matter**, *Phys. Rev. X* **16**, 031061 (2026), *equal contribution. Featured in *Physics* magazine and on the cover of *Phys. Rev. X*.

- Trained classifier to distinguish particle classes with nonreciprocal interactions, enabling analyses of previously experimentally unobserved nonequilibrium states. **Skills:** `scikit-learn`, `scipy`, `numpy`, `pandas`

Change-point detection with convergence guarantee: [code]

- Developed `scikit-learn`-style Python interface for detecting change points in noisy time series, implementing the current state-of-the-art dynamic-programming-based algorithm.

GRANTS AND LEADERSHIP

National Science Foundation Graduate Research Fellowship (GRFP, 2024, Physics)
National Defense Science and Engineering Graduate Fellowship (2024, Physics, declined for GRFP)
MIT Society of Physics Students and Sigma Pi Sigma: honor-society inductee (2022); President (2022)
Phi Kappa Theta – Massachusetts Eta: Pledgemaster (2021); sourced and fundraised donation of 12,000 surgical masks to Beth Israel Deaconess Medical Center during pandemic shortage as Service Chair (2019–2020).

SKILLS AND INTERESTS

Computing: `lightning`, `jax`, `pydantic`, `numpy`, `scipy`, `torch`, `scikit-learn`, `statsmodels`, `uv`, `ty`, `ruff`, `modal`
Natural language: Standard Chinese (native) · **Personal interests:** phonology, historical linguistics, ichthyology
Research interests: nonequilibrium statistical mechanics, time series, optimal transport, LLM expressivity